

Week 15 Self-Assessment

Correlation

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1 Instructions

This quiz applies everything from Weeks 12-14. You'll practice selecting and interpreting statistical tests, including p-values and effect sizes. This is the last new content before your Research Plan is due.

2 Part 1: Hypothesis Testing Logic

2.1 Question 1: Null Hypothesis

The null hypothesis (H_0) typically predicts:

- (A) A significant effect
- (B) No effect or no difference
- (C) The direction of the effect
- (D) That the study will be successful

Why do we test the null hypothesis rather than directly testing our experimental hypothesis?

- (A) The null is easier to understand
- (B) We can calculate the probability of our data IF the null were true
- (C) The experimental hypothesis is too complex
- (D) It's just tradition

2.2 Question 2: p-Values

A p-value tells us:

- (A) The probability that the null hypothesis is true
- (B) The probability that our hypothesis is correct
- (C) The probability of obtaining our results (or more extreme) if the null hypothesis were true
- (D) The effect size

If $p = .03$, this means:

- (A) There's a 3% chance our hypothesis is wrong
- (B) There's a 3% chance of getting these results if there were no real effect

- (C) The effect size is 3%
- (D) 3% of participants showed the effect

2.3 Question 3: Statistical Significance

We typically reject the null hypothesis when:

- (A) $p > .05$
- (B) $p < .05$
- (C) $p = .05$
- (D) p is any value

A result with $p = .08$ is:

- (A) Statistically significant
 - (B) Not statistically significant at $\alpha = .05$
 - (C) Definitely not a real effect
 - (D) A large effect
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3 Part 2: t-Tests Applied

3.1 Question 4: Choosing the Right t-Test

Different participants in each condition →

- (A) Independent t-test
- (B) Paired t-test
- (C) One-sample t-test
- (D) Pearson's r

Same participants measured twice →

- (A) Independent t-test
- (B) Paired t-test
- (C) One-sample t-test
- (D) Pearson's r

3.2 Question 5: DataLab t-Test Selection

Room A vs Room B reaction times (different students in each room):

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) Chi-square

Pre-mood vs Post-mood ratings (same students before and after):

- (A) Independent t-test
- (B) Paired t-test

- (C) Pearson's r
- (D) Chi-square

3.3 Question 6: t-Test Assumptions

Which assumption applies to BOTH independent and paired t-tests?

- (A) Equal group sizes
- (B) Data should be approximately normally distributed
- (C) Participants must be matched on characteristics
- (D) Correlation between variables

The independent t-test additionally assumes:

- (A) Homogeneity of variance (similar SDs in both groups)
 - (B) Repeated measures
 - (C) Perfect normal distribution
 - (D) Large effect size
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4 Part 3: Effect Size

4.1 Question 7: Why Effect Size Matters

Effect size tells us:

- (A) Whether the result is significant
- (B) The p-value
- (C) The magnitude or practical importance of the difference
- (D) The sample size needed

A statistically significant result with a tiny effect size means:

- (A) The finding is very important
- (B) The effect is real but may not be practically meaningful
- (C) We should ignore the result
- (D) The sample was too small

4.2 Question 8: Cohen's d Interpretation

Cohen's $d = 0.2$ is considered:

- (A) Small
- (B) Medium
- (C) Large
- (D) Very large

Cohen's $d = 0.5$ is considered:

- (A) Small
- (B) Medium

- (C) Large
- (D) Very large

Cohen's $d = 0.8$ is considered:

- (A) Small
- (B) Medium
- (C) Large
- (D) Very large

4.3 Question 9: Effect Size Calculation

Cohen's d is calculated as:

- (A) Mean divided by SD
 - (B) Difference between means divided by pooled SD
 - (C) SD divided by mean
 - (D) Sample size divided by variance
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5 Part 4: Interpreting Output

5.1 Question 10: Reading Results

A t-test output shows: $t(48) = 2.35$, $p = .023$, $d = 0.67$

Is this result statistically significant at $\alpha = .05$? TRUE / FALSE

The effect size is:

- (A) Small
- (B) Medium
- (C) Large
- (D) Cannot tell

The degrees of freedom (df) are: __

5.2 Question 11: APA Reporting

The correct APA format for reporting a t-test is:

- (A) $t = 2.35$, significant
- (B) $t(48) = 2.35$, $p = .023$, $d = 0.67$
- (C) t-test was significant at .05 level
- (D) The groups were different ($t = 2.35$)

5.3 Question 12: Correlation Reporting

A correlation analysis shows: $r(38) = .45$, $p = .003$

This represents a

- (A) Weak

- (B) Moderate to strong
- (C) Perfect
- (D) Non-significant

positive relationship.

The result IS statistically significant. TRUE / FALSE

6 Part 5: Complete Test Selection

6.1 Question 13: Design to Test Matching

Fill in the correct test for each scenario:

Comparing **two groups of different people**: _____

Comparing the **same people at two time points**: _____

Examining the **relationship between two continuous variables**: _____

6.2 Question 14: Non-Parametric Alternatives

When data violate normality assumptions:

Independent t-test →

- (A) Mann-Whitney U
- (B) Wilcoxon signed-rank
- (C) Spearman's rho
- (D) Chi-square

Paired t-test →

- (A) Mann-Whitney U
- (B) Wilcoxon signed-rank
- (C) Spearman's rho
- (D) Chi-square

Pearson's r →

- (A) Mann-Whitney U
 - (B) Wilcoxon signed-rank
 - (C) Spearman's rho
 - (D) Chi-square
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7 Part 6: MCQ Exam Practice

7.1 Investigation 1

A researcher tests whether a mindfulness intervention reduces stress. 30 participants are randomly assigned to either a mindfulness group or a control group. Stress is measured after 4 weeks.

1. **Design:**

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

2. **IV:**

- (A) Group (mindfulness vs control)
- (B) Stress level
- (C) Time (4 weeks)
- (D) No IV

3. **DV:**

- (A) Group assignment
- (B) Stress level
- (C) Mindfulness practice
- (D) Participant age

4. **Appropriate test:**

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) Mann-Whitney U

5. **Results: $t(28) = 2.89$, $p = .007$, $d = 1.08$. This is:**

- Significant? TRUE / FALSE
- Effect size:
 - (A) Small
 - (B) Medium
 - (C) Large
 - (D) Cannot determine

6. **The correct conclusion is:**

- (A) The mindfulness group had significantly lower stress than controls
- (B) There was no significant difference
- (C) The control group had lower stress
- (D) We cannot draw any conclusions

7.2 Investigation 2

A researcher measures students' confidence before and after receiving exam feedback.

1. **Design:**

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

2. **IV:**

- (A) Time (before vs after feedback)
- (B) Confidence level
- (C) Exam grade
- (D) No IV

3. **DV:**

- (A) Time point
- (B) Confidence level
- (C) Feedback content
- (D) Student ID

4. **Appropriate test:**

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) Wilcoxon signed-rank

5. **Results: $t(24) = -3.45$, $p = .002$, $d = 0.69$. The negative t-value means:**

- (A) The result is not significant
- (B) Confidence decreased (before > after)
- (C) The calculation is wrong
- (D) The effect size is negative

7.3 Investigation 3

A researcher examines whether study hours predict exam scores in 60 students.

1. **Design:**

- (A) Between-subjects
- (B) Within-subjects
- (C) Matched pairs
- (D) Correlational

2. **IV:**

- (A) Study hours

- (B) Exam scores
- (C) Number of students
- (D) No IV

3. Appropriate test:

- (A) Independent t-test
- (B) Paired t-test
- (C) Pearson's r
- (D) ANOVA

4. Results: $r(58) = .52, p < .001$. This represents:

- (A) A weak relationship
- (B) A moderate to strong positive relationship
- (C) No relationship
- (D) A negative relationship

5. Can we conclude studying CAUSES better grades? TRUE / FALSE

8 Summary Checklist

Before moving on, make sure you can:

- ☐ Explain the logic of hypothesis testing (null hypothesis, p-value)
- ☐ Interpret p-values correctly (NOT the probability the null is true)
- ☐ Select the appropriate test for any design (independent t, paired t, Pearson's r)
- ☐ Identify non-parametric alternatives
- ☐ Interpret and report effect sizes (Cohen's d, r)
- ☐ Read and interpret statistical output
- ☐ Report results in APA format

! Research Plan Due Next Week

Your Research Plan is due Week 16. You need to specify your hypothesis, design, IV, DV, and statistical test. This quiz covers all the content you need!